

3	(a) sum of potential energy and kinetic energy of atoms/molecules/particles reference to random (distribution)	M1 A1	[2]
(b)	(i) as lattice structure is 'broken'/bonds broken/forces between molecules reduced (not molecules separate) no change in kinetic energy, potential energy increases internal energy increases	B1 M1 A1	[3]
	(ii) <i>either</i> molecules/atoms/particles move faster/ $\langle c^2 \rangle$ is increasing <i>or</i> kinetic energy increases with temperature (increases) no change in potential energy, kinetic energy increases internal energy increases	B1 M1 A1	[3]
4	(a) (i) as r decreases, energy decreases/work got out (due to)	M1	